

Problem

Given a string S, find the longest border of the string.

A border is a non-empty substring that is both a prefix and a suffix of the string, but it must be shorter than the entire string.

S = "ababa"

Prefixes: "a", "ab", "aba", "abab" Suffixes: "a", "ba", "aba", "baba"

The longest common prefix and suffix is "aba".

Therefore, the answer is:

Submissions

Input Format

A single line containing the string S.

Constraints

$1 \leq |S| \leq 10^5$ S contains lowercase English letters.

Leaderboard

Output Format

Print the longest border of S.

Sample Input 0

ababa

Discussions

Sample Output 0

aba

Sample Input 1

aaaa

Sample Output 1

aaa

Change Theme Language Java 15

```
1 import java.io.*;
2
3 public class Solution {
4     public static void main(String[] args) throw
5         BufferedReader br = new BufferedReader(n
6         String s = br.readLine().trim();
7         int n = s.length();
8         int[] lps = new int[n];
9         int len = 0;
10        int i = 1;
11        while (i < n) {
12            if (s.charAt(i) == s.charAt(len)) {
13                len++;
14                lps[i] = len;
15                i++;
16            } else if (len > 0) {
17                len = lps[len - 1];
18            } else {
19                lps[i] = 0;
20                i++;
21            }
22        }
23
24        int borderLength = lps[n - 1];
25
```

Line: 5 Col: 82

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☐ Test against custom input

Test case 0

Compiler Message

Test case 1

Success